

First name: *Zach*Last name: *McEwen*Student ID: *8768637**10*

Name and Student ID must be filled in for this page to be graded.

Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

$$\phi = \frac{\pi}{6}$$

$$dA = (h_\rho d\rho)(h_\theta d\theta)$$

$$= d\rho \cdot \rho \sin \phi d\theta$$

$$= \frac{1}{2} \rho d\rho d\theta$$

$$= \int_0^2 \int_0^{2\pi} \frac{1}{2} \rho d\theta d\rho$$

$$= \frac{1}{2} \int_0^2 \rho d\rho \int_0^{2\pi} d\theta = \frac{1}{2} \cdot 2\pi \cdot \left. \frac{\rho^2}{2} \right|_0^2 = 2\pi \quad \checkmark$$

□

First name: Zach

Last name: McEwen

Student ID: 8768637

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Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$$dA = (h_\theta d\theta)(h_z dz)$$

$$= r d\theta dz = 2 d\theta dz$$

$$= \int_0^2 \int_0^{2\pi} 2 d\theta dz$$

$$= 2 \int_0^2 dz \int_0^{2\pi} d\theta$$

$$= 2 \cdot 2 \cdot 2\pi = 8\pi$$



□

First name: AMY

Last name: WENTZEL

Student ID: 8820998

10

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Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

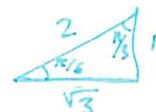
$$\phi = \frac{\pi}{6}, \quad d\phi = 0; \quad ds^2 = d\rho^2 + \rho^2 \sin^2 \phi d\theta^2 + \cancel{\rho^2 d\phi^2}$$

$$ds = d\rho + \rho \sin \phi d\theta$$

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \rho \leq 2$$

$$\therefore dA = \rho \sin \phi d\rho d\theta$$



$$SA = \int_0^2 \int_0^{2\pi} \rho \sin\left(\frac{\pi}{6}\right) d\theta d\rho, \quad \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$SA = \frac{1}{2} \int_0^2 \rho \theta \Big|_0^{2\pi} d\rho$$

$$SA = \frac{2\pi}{2} \int_0^2 \rho d\rho$$

$$SA = \pi \left(\frac{1}{2} \rho^2 \right) \Big|_0^2$$

$$SA = \frac{\pi}{2} (2)^2$$

$$SA = \frac{4\pi}{2}$$

$$\therefore SA = 2\pi$$

□

First name: AMY

Last name: WENTZELL

Student ID: 8820998

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Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$$r = 2, \quad dr = 0$$

$$ds^2 = \cancel{dr^2} + r^2 d\theta^2 + dz^2$$

$$ds = r d\theta + dz$$

$$0 \leq z \leq 2,$$

$$SA = r d\theta dz$$

$$0 \leq \theta \leq 2\pi,$$

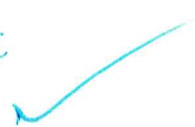
$$SA = \int_0^2 \int_0^{2\pi} 2 \, d\theta \, dz$$

$$SA = 2 \int_0^2 \theta \Big|_0^{2\pi} dz$$

$$SA = 2 \left(2\pi z \Big|_0^2 \right)$$

$$SA = 2 (4\pi)$$

$$\therefore SA = 8\pi$$



□

First name: Kyle

Last name: Dick

Student ID: 7206287

10

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Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

$$dA = (1)(\rho \sin \phi) d\theta d\rho$$

$$= \rho \sin(\pi/6) d\rho d\theta$$

$$= \frac{\rho}{2} d\rho d\theta$$

$$= \int_0^{2\pi} \int_0^2 \rho d\rho d\theta = \int_0^{2\pi} \left[\frac{\rho^2}{2} \right]_0^2 d\theta = \int_0^{2\pi} \frac{2^2}{2} d\theta = \int_0^{2\pi} 2 d\theta = 2 \int_0^{2\pi} 1 d\theta = 2 \left[\theta \right]_0^{2\pi} = 2(2\pi) = 4\pi$$

$$= 2\pi$$



□

First name: Kyle

Last name: Dick

Student ID: 7206287.

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Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution. $r = 2.$ $dA = r \, d\theta \, dz$

$$= 2 \, d\theta \, dz$$

$$= \int_0^2 \int_0^{2\pi} 2 \, d\theta \, dz = 2 \int_0^2 dz \int_0^{2\pi} d\theta = 2(2)(2\pi) = 8\pi.$$

□

First name: *Hayden Foster*

Last name:

Student ID: *8823130**10*

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Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

$$\phi = \text{Const} \rightarrow d\phi = 0 \quad \text{so} \quad dA = \rho \sin \phi d\rho d\theta$$

$$0 \leq \rho \leq 2$$

$$0 \leq \theta \leq 2\pi$$

Circle

$$A = \int_0^{2\pi} \int_0^2 \rho \sin \phi d\rho d\theta$$

$$\int_0^2 \rho \underbrace{\sin \frac{\pi}{6}}_{\frac{1}{2}} d\rho = \frac{1}{2} \int_0^2 \rho d\rho = \frac{\rho^2}{4} \Big|_0^2 = 1$$

$$\int_0^{2\pi} 1 d\theta = \theta \Big|_0^{2\pi} = 2\pi$$

~~$$A = 2\pi$$~~

$$A = 2\pi$$

□

First name: Hallden Foster

Last name:

Student ID: 8823130

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Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$r = \text{const} \rightarrow dr = 0$, $dA = r d\theta dz$
 $0 \leq z \leq 2$
 $0 \leq \theta \leq 2\pi$
 Circle
 $A = \int_0^2 \int_0^{2\pi} r d\theta dz$
 $2 \int_0^{2\pi} d\theta = 2\theta \Big|_0^{2\pi} = 4\pi$
 $\int_0^2 4\pi dz = 4\pi z \Big|_0^2 = 8\pi$
 $A = 8\pi$

First name: Coulter

Last name: Robb

Student ID: 4751065

10

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Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution. $\phi = \text{const}$ $\rho \sin(\phi) d\rho d\theta$

$$0 \leq \theta \leq 2\pi$$

$$\frac{1}{2} \rho d\rho d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \int_0^2 \rho d\rho d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \left[\frac{\rho^2}{2} \right]_0^2 d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} 2 d\theta$$

$$= [\theta]_0^{2\pi}$$

$$= 2\pi$$

□

First name: Coulter

Last name: Ralph

Student ID: 8751065

Name and Student ID must be filled in for this page to be graded.

Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution. $r = \text{const}$ $0 \leq \theta \leq 2\pi$

$r d\theta dz$

$$= \int_0^{2\pi} \int_0^2 r d\theta dz$$

$$= 2 \int_0^{2\pi} [\theta]_0^2 dz$$

$$= 4 \int_0^{2\pi} dz$$

$$= 4 [z]_0^{2\pi}$$

$$= 8\pi$$


□

First name: JOSHUA

Last name: DIPLOMA

Student ID: 7897160

10

Name and Student ID must be filled in for this page to be graded.

Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$



[5 marks]

Solution.

$$\phi = \text{const}$$

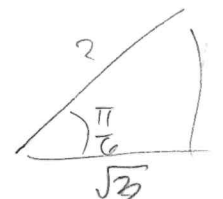
$$dA = \rho \sin \phi \, d\rho \, d\theta$$

$$\int dA = \sin \phi \int_0^2 \rho \, d\rho \int_0^{2\pi} d\theta$$

$$A = \sin\left(\frac{\pi}{6}\right) \left(\frac{2^2}{2}\right) (2\pi)$$

$$A = \left(\frac{1}{2}\right) (2) 2\pi$$

$$\boxed{A = 2\pi}$$



□

First name: JOSHUA

Last name: DIPLOMA

Student ID: 789796

Name and Student ID must be filled in for this page to be graded.

Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$$r = \text{const} = 2$$

$$0 \leq z \leq 2$$

$$dA = r \, d\theta \, dz$$

$$\int dA = r \int d\theta \, dz$$

$$A = (2) \int_0^{2\pi} d\theta \int_0^2 dz$$

$$A = (2)(2\pi)(2)$$

$$A = 8\pi$$



□

First name: Alan

Last name: Hosseinpour mogadam

Student ID: 8825132

10

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Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution. using the surface area over the flat region with constant $\phi = \frac{\pi}{6}$ we have

$$dA = h_\rho h_\theta d\rho d\theta \Rightarrow \text{sub in lamé coefficients}$$

$$\hookrightarrow dA = (1)(\rho \sin \phi) d\rho d\theta \Rightarrow dA = \rho \sin\left(\frac{\pi}{6}\right) d\rho d\theta$$

$$\hookrightarrow dA = \rho \left(\frac{1}{2}\right) d\rho d\theta \Rightarrow \int_0^{2\pi} \int_0^2 \frac{1}{2} \rho d\rho d\theta$$

we have

$$0 \leq \rho \leq 2$$

$$\hookrightarrow \frac{1}{2} \int_0^{2\pi} d\theta \int_0^2 \rho d\rho$$

$$0 \leq \theta \leq 2\pi$$

$$\hookrightarrow dA = \frac{1}{2} (2\pi - 0) \left(\frac{\rho^2}{2} \Big|_0^2 \right) \Rightarrow \frac{1}{2} (2\pi) \left(\frac{2^2 - 0}{2} \right)$$

$$\hookrightarrow dA = 2\pi \left(\frac{4}{2} \right) \Rightarrow \boxed{dA = 2\pi} \quad \therefore \text{surface area using spherical coordinates is } 2\pi \quad \square$$

First name: Alan Moscinpour

Last name:

Student ID: 8825132

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Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$r = 2$ constant

Therefore, we have $dA = h_\theta h_z d\theta dz$

$$\hookrightarrow dA = (r)(1) d\theta dz \Rightarrow dA = r d\theta dz$$

$$\hookrightarrow dA = 2 d\theta dz$$

$$0 \leq z < 2$$

$$0 \leq \theta \leq 2\pi$$

$$\rightarrow \int_0^{2\pi} \int_0^2 2 dz d\theta \Rightarrow 2 \int_0^{2\pi} d\theta \int_0^2 dz$$

$$\hookrightarrow 2(2\pi - 0)(2 - 0) = \boxed{8\pi = \text{surface area}}$$

□

Student ID: 8823349

10

Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

Solution.

$$\phi = \frac{\pi}{6}$$

$$0 \leq p \leq 2$$

$$0 \leq \theta \leq 2\pi$$

$$\int_0^{2\pi} \int_0^2 \rho \sin \phi \, d\rho \, d\theta$$

$$* \sinh\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$= \int_0^{2\pi} \int_0^2 p \sin\left(\frac{\pi}{6}\right) d\rho d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \int_0^2 \rho \, d\rho \, d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \frac{p^2}{2} \Big|_0^2 d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} 2 \, d\theta$$

$$= \frac{1}{2} [2\theta]_0^{2\pi}$$

$$= \frac{1}{2} [4\pi]$$

$$= 2\pi$$

First name: David

Last name: Toma

Student ID: 8823349

Name and Student ID must be filled in for this page to be graded.

Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$$R = 2$$

$$0 \leq z \leq 2$$

$$0 \leq \theta \leq 2\pi$$

dA  cylindrical

$$\int_0^{2\pi} \int_0^2 R \, dz \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 2 \, dz \, d\theta$$

$$= \int_0^{2\pi} 2z \Big|_0^2 \, d\theta$$

$$= \int_0^{2\pi} 4 \, d\theta$$

$$= 4\theta \Big|_0^{2\pi}$$

$$= 8\pi$$

□

First name: Stuart

Last name: Murray

Student ID: 8864228

10

Name and Student ID must be filled in for this page to be graded.

Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

$$\phi = \text{const} = \frac{\pi}{6}$$

$$= \int_0^{2\pi} \int_0^2 \frac{1}{2} \rho \, d\rho \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{1}{2} \rho^2 \right]_0^2 d\theta$$

$$= \int_0^{2\pi} 2 \, d\theta$$

$$= 2\pi$$

✓

$$\begin{aligned} dA &= (1)(\rho \sin \theta)(d\rho d\theta) \\ &= \rho \sin\left(\frac{\pi}{6}\right) d\rho d\theta \\ &= \frac{1}{2} \rho d\rho d\theta \end{aligned}$$

✓

□

First name: *Stuart*Last name: *Murray*Student ID: *8854228*

Name and Student ID must be filled in for this page to be graded.

Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$$r = \text{const} = 2$$

$$\begin{aligned} dA &= (1)(r) d\theta dz \\ &= 2 d\theta dz \end{aligned}$$

$$= \int_0^{2\pi} \int_0^2 2 dz d\theta$$

$$= \int_0^{2\pi} 2z \Big|_0^2 d\theta$$

$$= \int_0^{2\pi} 4 d\theta$$

$$= 8\pi$$

□

First name:

Last name:

Student ID:

Jesse
Russell
2240

10

Name and Student ID must be filled in for this page to be graded.

Problem 1. Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 marks]

Solution.

$$\phi = \frac{\pi}{6} = c$$

$$dA = (h_\rho d\rho)(h_\phi d\phi)$$

$$dA = \rho \sin \phi d\rho d\phi$$

$$= \rho \sin\left(\frac{\pi}{6}\right)$$

$$dA = \frac{1}{2} \rho d\rho d\theta$$

Volume

$$= \int_0^{2\pi} \int_0^2 \frac{1}{2} \rho d\rho d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} d\theta \int_0^2 \rho d\rho$$

$$= \frac{1}{2} (2\pi) \left(\frac{\rho^2}{2}\right) \Big|_0^2$$

$$= \pi \left(\frac{2^2}{2} - \frac{0^2}{2}\right)$$

$$= 2\pi$$



□

First name:

Last name:

Student ID:

Jesse
Russell
2240

Name and Student ID must be filled in for this page to be graded.

Problem 2. Using double integrals and the **Cylindrical coordinates**, compute the surface area of a cylinder of radius 2, with $0 \leq z \leq 2$.

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

[5 marks]

Solution.

$$r = 2 \rightarrow \text{constant}$$

$$dA = (h_\theta d\theta) (h_z dz)$$

$$= r d\theta dz$$

~~SA~~ Surface Area

$$= \int_0^{2\pi} \int_0^2 2 dz d\theta$$

$$= 2 \int_0^{2\pi} d\theta \int_0^2 dz$$

$$= 2 (2\pi) (2-0)$$

$$= 8\pi \quad \checkmark$$

□